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基于案例推理系统中用于检索的相似性度量

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SIMILARITY MEASURES FOR RETRIEVAL IN CASE-BASED REASONING SYSTEMS

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Case-based reasoning (CBR) is one of the emerging paradigms for designing intelligent systems. Retrieval of similar cases is a primary step in CBR, and the similarity measure plays a very important role in case retrieval. Sometimes CBR systems are called similarity searching systems, the most important characteristic of which is the effectiveness of the similarity measure used to quantify the degree of resemblance between a pair of cases. This article focuses on the similarity measuring methods for CBR and comprises two parts. The first part reviews the existing methods for measuring similarity in the literature based on more than 100 CBR project studies and some general similarity measures seen in other applications. In the second part, a hybrid similarity measure is proposed for comparing cases with a mixture of crisp and fuzzy features. Its application to the domain of failure analysis is illustrated.

Case-based reasoning (CBR) is one of the emerging paradigms for designing intelligent systems. It is an alternative approach to rule-based systems. The advantages of the CBR approach are many, the most important of which is that it parallels the way people solve problems in various situations. It overcomes the drawbacks of rule-based systems with the following advantages:

- CBR does not require an explicit domain knowledge model, and the problem of knowledge elicitation bottleneck can be avoided;
- the main implementation of CBR is reduced to identify significant attributes;
- large volumes of historical cases can be efficiently managed by advanced database techniques; and
- by obtaining new cases, CBR can learn and improve its performance, and maintenance is easier.

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基于案例推理系统中相似性检索的度量方法

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Case-based reasoning (CBR) is one of the emerging paradigms for designing intelligent systems. Retrieval of similar cases is a primary step in CBR, and the similarity measure plays a very important role in case retrieval. Sometimes CBR systems are called similarity searching systems, the most important characteristic of which is the effectiveness of the similarity measure used to quantify the degree of resemblance between a pair of cases. This article focuses on the similarity measuring methods for CBR and comprises two parts. The first part reviews the existing methods for measuring similarity in the literature based on more than 100 CBR project studies and some general similarity measures seen in other applications. In the second part, a hybrid similarity measure is proposed for comparing cases with a mixture of crisp and fuzzy features. Its application to the domain of failure analysis is illustrated.

基于案例的推理（CBR）是设计智能系统的一种新兴范式，它是一种有别于基于规则系统的替代方法。基于案例的推理方法具有诸多优势，其中最为重要的一点是，其思维方式与人类在各种情境中解决问题的方式高度一致。该方法克服了基于规则系统的诸多缺陷，并具备以下优点：

它无需显式的领域知识模型，可避免基于案例推理中存在的知识获取瓶颈问题；基于案例推理的主要实现过程简化为识别关键属性；借助先进的大型数据库技术，可以高效管理海量历史案例；并且在获取新案例的过程中，基于案例推理能够自主学习并提升其性能，同时维护也更为便捷。

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Typically, there are five steps in CBR: presentation of a new case, retrieval of the most similar cases from the case base, adaptation of the selected cases, validation of the current solution given by CBR, and updating of the system by adding the verified solution to the case base. The corresponding methodology for developing a CBR system should closely follow these steps. Among them, case retrieval is a primary step in CBR applications. Without effective case retrieval, a CBR system cannot do anything. Of course, enough cases should be first saved in the case memory, so that a similar one can be retrieved.

The similarity measure used to quantify the degree of resemblance between a pair of cases plays a very important role in case retrieval. Therefore CBR systems are sometimes called similarity searching systems. Because of the importance of similarity measure in CBR, this article focuses on the similarity measuring methods. The article is divided into two parts. Part 1 reviews the existing methods for measuring similarity in the open literature based on more than 100 CBR project studies and some general similarity measures seen in other applications including those using crisp and fuzzy set theories. In part 2, a hybrid similarity measure is proposed to compare cases having a mixture of crisp and fuzzy features. Its application to the domain of failure analysis is demonstrated. The conclusions are given in the final section.

SIMILARITY MEASURES FOR CBR

There are two major case retrieval approaches:

- One is the distance-based or computational approach, which calculates the distance between cases from a number of objects constituting the cases. The most similar case is determined by the evaluation of a similarity measure.
- Second is the indexing or representational approach. The similar case is coded into the structure of the case base itself. The cases are connected by indexing structures. The indexing structure can be traversed to search the similar case. Another situation is based on the graphical appearance of the cases, which compares abstracted bitmaps of design pictures.

Note that the above two approaches might be used in combination. Our similarity measures discussed below will focus on the first approach because it is the most widely used in CBR applications.

Basic CBR Similarity Measures

Commonly, CBR systems use the inverse of weighted normalized Euclidian distance or Hamming distance as the similarity measure. That is,

通常，基于案例的推理（CBR）包含五个步骤：呈现新案例、从案例库中检索最相似的案例、对所选案例进行适应性调整、验证CBR给出的当前解决方案，以及通过将经验证的解决方案添加至案例库来更新系统。相应的CBR系统开发方法应严格遵循这些步骤。其中，案例检索是CBR应用中的关键环节；若缺乏有效的案例检索，CBR系统将无法正常运行。当然，首先需要在案例存储中保存足够数量的案例，以便能够检索到与之相似的案例。

用于量化两例之间相似程度的相似性度量在案例检索中起着至关重要的作用。因此，案例推理系统有时也被称为相似性搜索系统。鉴于相似性度量在案例推理中的重要性，本文重点探讨相似性度量方法。全文分为两部分：第一部分基于百余项案例推理项目研究，以及在其他应用领域（包括采用经典集理论和模糊集理论的方法）中常见的若干通用相似性度量，对现有文献中的相似性度量方法进行了综述；第二部分提出了一种混合型相似性度量，用于比较同时包含精确特征与模糊特征的案例，并展示了其在故障分析领域的应用。最后，在结论部分给出了相关总结。

案例推理的相似性度量

目前主要有两种案例检索方法：

一种是基于距离或计算的方法，该方法通过计算构成案例的多个对象之间的距离来确定最相似的案例；最相似的案例由相似度度量的评估结果决定。另一种是索引或表示方法，其中最相似的案例被编码到案例库的结构本身中；案例之间通过索引结构相互关联，可通过遍历索引结构来搜索最相似的案例。还有一种方法则基于案例的图形外观，通过对设计图纸的抽象位图进行比较。

需要注意的是，上述两种方法可以结合使用。我们下文所讨论的相似性度量将主要聚焦于第一种方法，因为该方法在案例推理应用中最为广泛。

基于案例的推理基本相似性度量

通常，基于案例的推理系统将加权归一化欧几里得距离或汉明距离的倒数作为相似性度量。也就是说，

$$\mathrm{SIM}(\mathbf{X}, \mathbf{Y}) = 1 - \mathrm{DIST}(\mathbf{X}, \mathbf{Y}) = 1 - \sqrt{\sum_i w_i^2 \, \mathrm{dist}^2(x_i, y_i)} \tag{1a}$$

use Euclidian distance, and

$$\mathrm{SIM}(\mathbf{X}, \mathbf{Y}) = 1 - \mathrm{DIST}(\mathbf{X}, \mathbf{Y}) = 1 - \sum_i w_i \, \mathrm{dist}(x_i, y_i) \tag{1b}$$

use Hamming distance, where weight w_i is normalized for denoting the importance of the ith attribute and $i = 1, 2, \ldots, n$, n being the number of attributes in the cases. The normalized $\mathrm{dist}(x_i, y_i)$ is often represented as

$$\mathrm{dist}(x_i y_i) = |x_i - y_i| / |\max_i - \min_i|$$

For numerical attributes, $\max_i$ and $\min_i$ are the maximum and minimum values of the ith attributes, respectively. For symbolic attributes, if $x_i = y_i$, $\mathrm{dist}(x_i, y_i) = 0$; otherwise, $\mathrm{dist}(x_i, y_i) = 1$. Equation (1) shows that when DIST = 0, SIM attains the maximum value 1, and this means two cases are identical. When DIST = 1, the SIM has a minimum value 0, meaning two cases are completely different.

Another widely used similarity measure is based on the ratio mode proposed by Tversky (1977):

$$\mathrm{SIM}(\mathbf{X}, \mathbf{Y}) = \frac{\alpha \times \mathrm{common}}{\alpha \times \mathrm{common} + \beta \times \mathrm{different}} \tag{2}$$

where *common* (*different*) represents the number of attributes whose values are classified as similar (dissimilar) between the old case and the new case. Two values are classified as similar (dissimilar) if their similarity is above (below) some threshold; α and β are the corresponding weights for *common* and *different*.

Hunt et al. (1995) proposed another similarity measure based on metaphors taken from the human immune system. The measure assumes the new case to be a new antigen and the old case to be an old antibody. The match score is calculated as

$$\mathrm{SIM}(\mathbf{X}, \mathbf{Y}) = \mathrm{count} + \sum_i 2^{l_i} \tag{3}$$

where *count* is the number of fields that match between the antigen and the antibody, and l_i is the length of each region consisting of two or more matches. In this measure, case attributes are coded as strings in some specific

$$\mathrm{SIM}(\mathbf{X}, \mathbf{Y}) = 1 - \mathrm{DIST}(\mathbf{X}, \mathbf{Y}) = 1 - \sqrt{\sum_i w_i^2 \, \mathrm{dist}^2(x_i, y_i)} \tag{1a}$$

使用欧几里得距离，并

$$\mathrm{SIM}(\mathbf{X}, \mathbf{Y}) = 1 - \mathrm{DIST}(\mathbf{X}, \mathbf{Y}) = 1 - \sum_i w_i \, \mathrm{dist}(x_i, y_i) \tag{1b}$$

采用汉明距离，并对权重进行归一化处理，w_i以表征第i个属性的重要性，其中i取值为1, 2, …, n，n为案例中属性的总数。归一化后的距离通常记为dist(x_i, y_i)。

$$\mathrm{dist}(x_i, y_i) = |x_i - y_i| / |\max_i - \min_i|$$

对于数值型属性，$\max_i$和$\min_i$分别为第i个属性的最大值和最小值。对于符号型属性，若$x = y$，则；否则，由式(1)可知，当dist(x, y) = 0时，dist(x, y) = 1。当DIST = 0时，SIM取最大值1，此时表明两种情况完全相同；当DIST = 1时，SIM取最小值0，此时表明两种情况完全不相同。

另一种广泛使用的相似性度量基于Tversky（1977）提出的比率模式：

$$\mathrm{SIM}(\mathbf{X}, \mathbf{Y}) = \frac{\alpha \times \mathrm{common}}{\alpha \times \mathrm{common} + \beta \times \mathrm{different}} \tag{2}$$

其中，*common* (*different*) 表示在旧案例与新案例之间，其取值被归类为相似（不相似）的属性数量。若两者的相似度高于（低于）某一阈值，则将其取值归类为相似（不相似）；α 和 β 分别为 *common* 和 *different* 对应的权重。

Hunt 等人（1995）提出了一种基于人类免疫系统隐喻的相似性度量方法。该度量假设新案例为一种新的抗原，而旧案例则被视为一种旧的抗体。匹配得分的计算公式为：

$$\mathrm{SIM}(\mathbf{X}, \mathbf{Y}) = \mathrm{count} + \sum_i 2^{l_i} \tag{3}$$

其中，*count* 表示抗原与抗体之间匹配的®域数量，而 是由两个或更多 l_i 匹配构成的每个区域的长度。在该度量中，案例属性被编码为特定格式的字符串。

order. For example, two cases **X**, **Y** have matching score 13 as illustrated below:

new case (**X**):	Adventure	2498	2	Egypt	Car	14	March	*
old case (**Y**):	Adventure	3066	3	Egypt	Car	14	April	*
count	1	0	0	1	1	1	0	$1 \Rightarrow 5$

SIM(**X**, **Y**): $5 + 2^3 \Rightarrow 13$

Sebag and Schoenauer (1993) discussed a rule-based similarity measure. Let Ω be the problem domain. Given two examples (cases) **X** and **Y** in Ω, the similarity between them SIM(**X**, **Y**) was considered to be the number of rules that are fired by both or by none of **X** and **Y**. Each rule R_i learned from the case base is assigned a weight $w(R_i)$ depending upon the number of training examples it covers. Thus the similarity measure is defined as

$$\forall \mathbf{X}, \mathbf{Y} \in \Omega \qquad \mathrm{SIM}(\mathbf{X}, \mathbf{Y}) = \sum_i w(R_i). \tag{4}$$

Because case matching is not a symmetrical operation in some situations, Weber (1994) used the contrast model (Tversky, 1977), in which the similarity between a target case **X** (new case) and a base case **Y** (old case) was given by the following formula:

$$\mathrm{SIM}(\mathbf{X}, \mathbf{Y}) = \theta f(\mathbf{X} \cap \mathbf{Y}) - \alpha f(\mathbf{X} - \mathbf{Y}) - \beta f(\mathbf{Y} - \mathbf{X}) \tag{5}$$

where the intersection $(\mathbf{X} \cap \mathbf{Y})$ describes the set of attributes that are common to **X** and **Y**, and the complement sets (**X**-**Y**) and (**Y**-**X**) describe attributes observed only in the target case and attributes observed only in the base case, respectively. Here θ, α, β are the weighting parameters denoting different importance for each term. The f is some operator or algorithm to compute the matching score of the related sets. Pal and Campbell (1995) defined the mutual similarity coefficient for **X** and **Y** as

$$\mathrm{SIM}(\mathbf{X}, \mathbf{Y}) = (S_{lr} + S_{rl})/2 \tag{6}$$

where S_{lr} is the normalized association of **X** with respect to **Y** and S_{rl} is the normalized association of **Y** with respect to **X**. S_{lr} is defined as $|\mathbf{X} \cap \mathbf{Y}|/|\mathbf{X}|$ denoting the ratio of the number of attributes common to **X** and **Y** to the size of **X**. Similarly, S_{rl} is defined by replacing $|\mathbf{X}|$ with $|\mathbf{Y}|$.

Stanfill and Waltz (1986) computed the distance DIST between two cases **X** and **Y** as

$$\mathrm{DIST}(\mathbf{X}, \mathbf{Y}) = \sum_i w_i(x_i) \times \mathrm{diff}(x_i, y_i) \tag{7}$$

顺序。例如，两个案例X和Y的匹配得分均为13，如下所示：

new case (**X**):	Adventure	2498	2	Egypt	Car	14	March	*
old case (**Y**):	Adventure	3066	3	Egypt	Car	14	April	*
count	1	0	0	1	1	1	0	$1 \Rightarrow 5$

SIM(**X**, **Y**): $5 + 2^3 \Rightarrow 13$

Sebag和Schoenauer（1993）讨论了一种基于规则的相似性度量。设W为问题域，给定W中的两个实例（案例）**X**和**Y**，它们之间的相似性SIM(**X**, **Y**)被定义为同时适用于两者或均不适用于两者的规则数量。从案例库R中学习得到的每条规则均根据其所覆盖的训练实例$w(R_i)$的数量被赋予一个权重。因此，该相似性度量被定义为

$$\forall \mathbf{X}, \mathbf{Y} \in \Omega \qquad \mathrm{SIM}(\mathbf{X}, \mathbf{Y}) = \sum_i w(R_i). \tag{4}$$

由于在某些情况下，案例匹配并非对称操作，Weber（1994）采用了对比模型（Tversky，1977），其中目标案例X（新案例）与基准案例Y（旧案例）之间的相似度由如下公式给出：

$$\mathrm{SIM}(\mathbf{X}, \mathbf{Y}) = \theta f(\mathbf{X} \cap \mathbf{Y}) - \alpha f(\mathbf{X} - \mathbf{Y}) - \beta f(\mathbf{Y} - \mathbf{X}) \tag{5}$$

其中，交集（X ∩ Y）表示同时属于 X 和 Y 的属性集合，而补集（X - Y）和（Y - X）分别表示仅在目标案例中观察到的属性以及仅在基准案例中观察到的属性。此处，θ、α、β 为权重参数，用于表征各术语的不同重要性。f 则为用于计算相关集合匹配得分的某种算子或算法。Pal 和 Campbell（1995）将 X 和 Y 的互相似系数定义为

$$\mathrm{SIM}(\mathbf{X}, \mathbf{Y}) = (S_{lr} + S_{rl})/2 \tag{6}$$

其中，S_{lr} 是 X 相对于 Y 的归一化关联度，S_{rl} 是 Y 相对于 X 的归一化关联度。S_{lr} 定义为 n_{lr}/n_l，其中 n_{lr} 表示 X 与 Y 共有的属性数量，n_l 与 X 的大小之比。类似地，通过将 l 替换为 r 而定义 S_{rl}。

Stan® ll 和 Waltz（1986）将两个案例 X 和 Y 之间的距离 DIST 计算为

$$\mathrm{DIST}(\mathbf{X}, \mathbf{Y}) = \sum_i w_i(x_i) \times \mathrm{diff}(x_i, y_i) \tag{7}$$

The similarity between individual (symbolic) feature values was calculated using a value-difference-metric (VDM). The weight of the ith attribute with value x_i, $w_i(x_i)$, and the difference between two attribute values, $\text{diff}(x_i, y_i)$, are defined as follows:

$$w_i(x_i) = \sqrt{\sum_{c \in C} \left[\lambda(c, x_i) / \lambda(x_i) \right]^2} \tag{8}$$

$$\text{diff}(x_i, y_i) = \sum_{c \in C} \left(\frac{\lambda(c, x_i)}{\lambda(x_i)} - \frac{\lambda(c, y_i)}{\lambda(y_i)} \right)^2 \tag{9}$$

where $i = 1, \ldots, n$, n being the number of attributes, $\lambda(c, x_i)$ is the number of times that cases of class c in the case library have value x_i for the ith attribute, whereas $\lambda(x_i)$ is the same value summed over the set of all classes C. VDM assigns higher weights to attributes whose distribution of values across classes is highly skewed and greater differences to values whose corresponding sets of cases have highly disparate class distributions. Thus two cases are similar if they have attribute values whose respective projections on the training library have similar class distributions.

Enhancement of Basic CBR Similarity Measures

Case Typicality

CBR used for supporting human decision such as classification and planning implicitly expresses the knowledge in cases. It is natural that some cases contain more knowledge than others. The former are more typical cases in the case space. Agre (1995) proposed a method that considers case typicality, that is,

$$\text{DIST}(\mathbf{X}, \mathbf{Y}) = W_{\mathbf{Y}} \times \text{Dist}(\mathbf{X}, \mathbf{Y}) \tag{10}$$

where $\text{Dist}(\mathbf{X}, \mathbf{Y})$ is the distance between new case $\mathbf{X}$ and old case $\mathbf{Y}$ without considering the case typicality. $W_{\mathbf{Y}}$ is the weight of old case $\mathbf{Y}$. The weight of an old case is simply the reciprocal of its typicality. When more than one case belonging to different classes is found, the most typical one is preferred. The case typicality can be determined by the expert or by calculating the average distance between the tested case and selected typical cases.

Attribute Relevance

To reflect the attribute relevance, Cain et al. (1991) proposed an approach based on case explanation. They used a parameterized similarity function influenced by the explanations produced by the domain theory for

个体（符号型）特征值之间的相似性采用值差度量（VDM）进行计算。第i个属性的权重及其两个属性值之间的差异x_i、$w_i(x_i)$，记为：

$$w_i(x_i) = \sqrt{\sum_{c \in C}\left[\lambda(c, x_i)/\lambda(x_i)\right]^2} \tag{8}$$

$$\mathrm{diff}(x_i, y_i) = \sum_{c \in C}\left(\frac{\lambda(c, x_i)}{\lambda(x_i)} - \frac{\lambda(c, y_i)}{\lambda(y_i)}\right)^2 \tag{9}$$

其中，i 至 n、n 分别表示属性的数量；$\lambda(c$ 表示在案例库$_i$中，属于类别 c 的案例在第 i 个属性上取值的次数；而 x 则为对该属性在所有类别上的取值次数进行求和所得的结果 C。VDM 会为那些在不同类别间取值分布极度不均衡的属性赋予更高的权重，并对那些对应案例集在各类别中分布差异显著的取值给予更大的权重。因此，当两个案例的各属性取值在训练库中的投影所对应的类别分布相似时，即可认为这两个案例是相似的。

基本CBR相似度度量的增强

Case Typicality

案例推理用于支持诸如分类与规划等人类决策，其知识在案例中得以隐式表达。显然，某些案例所蕴含的知识量多于其他案例，前者在案例空间中属于更为典型的案例。Agre（1995）提出了一种考虑案例典型性的方法，即，

$$\mathrm{DIST}(\mathbf{X}, \mathbf{Y}) = W_Y \times \mathrm{Dist}(\mathbf{X}, \mathbf{Y}) \tag{10}$$

其中，Dist(X, Y) 表示在不考虑案例典型性的情况下，新案例 X 与旧案例 Y 之间的距离；为旧案例 Y 的权重。对于旧案例 W_Y，其权重即为其典型性的倒数。当发现多个属于不同类别的案例时，优先选择最典型的案例。案例的典型性可由专家确定，也可通过计算待测案例与所选典型案例之间的平均距离来确定。

Attribute Relevance

为反映属性相关性，Cain 等人（1991）提出了一种基于案例解释的方法。他们采用了一个受领域理论所生成的解释影响的参数化相似度函数。

each case in memory:

$$\mathrm{SIM}(\mathbf{X}, \mathbf{Y}) = \frac{\alpha \sum_i \mathrm{sim}(x_i, y_i) + \beta \sum_i \mathrm{relevance}(y_i) \times \mathrm{sim}(x_i, y_i)}{\alpha n + \beta \sum_i \mathrm{relevance}(y_i)} \tag{11}$$

where

$$\begin{aligned} \mathrm{sim}(x_i, y_i) &= 1 \qquad x_i = y_i \\ \mathrm{sim}(x_i, y_i) &= 0 \qquad x_i \neq y_i \\ \\ \mathrm{relevance}(y_i) &= 1 \qquad y_i \text{ is relevant} \\ \mathrm{relevance}(y_i) &= 0 \qquad y_i \text{ is irrelevant} \end{aligned}$$

Here y_i is a fact in the past problem and x_i is a fact in the new problem. A fact y_i is relevant if it is a premise for an explanation in the past case; α, β are the corresponding term weights. When β is set to zero, the measure ignores the relevance of facts, and a pure similarity-based match is performed.

Transforming Attributes

Different problems usually have different attributes whose expressing methods and measuring scales may be quite different. It is sometimes necessary to transform original problems and their attributes into equivalent ones in order to apply the CBR method to help in problem solving. To illustrate the equivalence of the learning power of symbolic and case-based methods, Wess and Globig (1993) studied the transformation of a simple symbolic learning algorithm (the version space algorithm) into an equivalent case-based variant. Symbolic learning approaches represent the concepts learned explicitly, while case-based approaches describe concepts implicitly by a pair (CB, SIM), i.e., by a measure of similarity SIM and a set of cases CB. In the version space algorithm, the main learning task is to distinguish between relevant and irrelevant attributes. An attribute value is called relevant if it is part of the target concept $\mathbf{X} = (x_1, \ldots, x_n)$. Function f_i is defined as

$$\begin{aligned} f_i(y_i) &= 1 \qquad x_i = y_i \\ f_i(y_i) &= 0 \qquad \text{otherwise} \end{aligned}$$

The functions f_i are then combined into one $f\colon U \rightarrow \{0, 1\}^n$, $f[(x_1, \ldots, x_n)] = [f_1(x_1), \ldots, f_n(x_n)]$. The distance between two cases $\mathbf{X}$ and $\mathbf{Y}$ is defined using the city-block metric as follows:

$$\mathrm{DIST}(\mathbf{X}, \mathbf{Y}) = |f_1(x_1) - f_1(y_1)| + \cdots + |f_n(x_n) - f_n(y_n)| \tag{12}$$

内存中的每个案例:

$$\mathrm{SIM}(\mathbf{X}, \mathbf{Y}) = \frac{\alpha \sum_i \mathrm{sim}(x_i, y_i) + \beta \sum_i \mathrm{relevance}(y_i) \times \mathrm{sim}(x_i, y_i)}{\alpha n + \beta \sum_i \mathrm{relevance}(y_i)} \quad (11)$$

其中

$$\mathrm{sim}(x_i, y_i) = 1 \qquad x_i = y_i$$
$$\mathrm{sim}(x_i, y_i) = 0 \qquad x_i \neq y_i$$

$$\mathrm{relevance}(y_i) = 1 \qquad y_i \text{ is relevant}$$
$$\mathrm{relevance}(y_i) = 0 \qquad y_i \text{ is irrelevant}$$

此处所述既为旧问题中的一个事实，亦为新问题中的一个事实。若某一事实是旧案例中某解释的前提，则该事实被视为相关；α、β、γ分别为相应的项权重。当β取零时，该度量将忽略事实的相关性，仅执行基于相似度的匹配。

Transforming Attributes

不同的问题通常具有不同的属性，这些属性的表达方式和度量尺度可能大相径庭。为了将案例推理方法应用于问题求解，有时有必要对原始问题及其属性进行等价变换。为说明符号学习方法与基于案例的学习方法在学习能力上的等价性，Wess 和 Globig（1993）研究了将一种简单的符号学习算法（版本空间算法）转化为与其等价的基于案例的变体。符号学习方法以显式方式表示所学概念，而基于案例的方法则通过一对要素（CB, SIM），即相似度度量 SIM 和案例集 CB，以隐式方式描述概念。在版本空间算法中，主要的学习任务是区分相关属性与无关属性。如果某个属性值属于目标概念的一部分，则称该属性值为相关属性……函数被定义为 X 5 (x_1, x)。

$$f_i(y_i) = 1 \qquad x_i = y_i$$
$$f_i(y_i) = 0 \qquad \text{otherwise}$$

随后，这些函数被组合为一个f: $U\{0, 1\}, \cdots, f[(x_1, x_n)]5\cdots$。两个样本$\mathbf{X}$和$\mathbf{Y}$之间的距离采用$[f_1(x_1), f(x)]$的城市街区度量来定义，具体如下:

$$\mathrm{DIST}(\mathbf{X}, \mathbf{Y}) = |f_1(x_1) - f_1(y_1)| + \cdots + |f_n(x_n) - f_n(y_n)| \quad (12)$$

Every change of the $f_i (i = 1, \ldots, n)$ causes a change of the underlying measure DIST. Case-based learning may be seen as a concept-forming task. For each intended concept, the function f_i is learned by some algorithm through selected positive and negative cases. Once the concept is learned, the function f_i and the CB can be used for classification.

Adaptability Guided Measure

When a CBR system is measured with a new case, it will retrieve the most similar case from its case base and, if necessary, adapt it to provide the desired solution. Smyth and Keane (1993) argued that the success of CBR is critically dependent upon the retrieval of a suitable case that is the easiest, of those available, to adapt. Assume that the case attributes are {Action, Vehicle, Direction, Destination, Speed}. Case **Y1** is {Move, Buggy, Forward, Tension-Reel, 1-speed}. Case **Y2** is {Lift, Lifter, Upward, Spool-stand, 2-speed}. Target case **X** is {Move, Buggy, Forward, Tension-Reel, 2-speed}. Measuring the semantic similarity between the target and candidates, case **Y1** is retrieved. However, according to the domain knowledge, modifying the speed of a case is far more difficult than modifying the other attributes. So, although the similarity between the target and case **Y1** is higher, case **Y2** is easier to adapt. One way to consider attribute adaptability is to assign an appropriate weight to the corresponding attribute. Smyth and Keane used a measure of the closeness between the target and candidates in the adaptation space. The retrieval space and adaptation space are linked using the adaptation knowledge. How to measure the attribute adaptability is a major issue in applying this concept.

Missing Attributes

In reality, the stored case and the target case will commonly contain some missing or null attribute values. These missing attribute values might appear in both cases or just one. Handling missing values is an important issue in similarity measure. Agre (1995) used the following method to estimate the distance of an attribute with missing values:

$$\text{dist}^2(x_i, y_i) = 1/L_i \times (1 - 1/L_i) \tag{13}$$

where L_i is the number of possible values of the ith attribute that is restricted to the symbolic type. Ricci and Avesani (1995) computed the similarity between two cases **X** and **Y** for normalized attribute i with missing value as follows:

$$\text{sim}(x_i, y_i) = 0.5 \qquad \text{if } x_i \text{ or } y_i \text{ is unknown} \tag{14}$$

每当……发生变化时，i都会引起底层度量DIST的变化。基于案例的学习可以被视为一项概念形成任务。对于每一个待学习的概念，该函数通过某种算法f，借助所选取的正例和反例进行学习。一旦概念被习得，该函数与CB即可用于分类。f_i

Adaptability Guided Measure

当一个基于案例的推理系统接收到一个新的案例时，它会从案例库中检索出与之最为相似的案例，并在必要时对该案例进行调整，以给出期望的解决方案。Smyth和Keane（1993）指出，基于案例的推理的成功在很大程度上取决于能否检索到一个合适的案例——在所有可用案例中，该案例是最易于调整的。假设案例的属性为{动作、车辆、方向、目的地、速度}。案例Y1为{移动，越野车，向前，张力卷筒，1档速度}，案例Y2为{提升，升降机，向上，线轴架，2档速度}，目标案例X为{移动，越野车，向前，张力卷筒，2档速度}。通过计算目标案例与候选案例之间的语义相似度，系统检索到了案例Y1。然而，根据领域知识，修改案例的速度远比修改其他属性更为困难。因此，尽管目标案例与案例Y1的相似度更高，但案例Y2却更易于调整。考虑属性可调整性的一种方法是为相应的属性赋予适当的权重。Smyth和Keane采用了一种在调整空间中衡量目标案例与候选案例接近程度的指标。检索空间与调整空间通过调整知识相互关联。如何度量属性的可调整性，是应用这一概念时的一个关键问题。

Missing Attributes

实际上，存储案例与目标案例通常会包含一些缺失或空的属性值。这些缺失的属性值可能同时出现在两个案例中，也可能仅出现在其中一个案例中。在相似性度量中，如何处理缺失值是一个重要问题。Agre（1995）采用如下方法来估计具有缺失值的属性之间的距离：

$$\mathrm{dist}^2(x_i, y_i) = 1/L_i \times (1 - 1/L_i) \tag{13}$$

其中，i表示第i个属性在被限制为符号型时的可能取值个数。Ricci和Avesani（1995）针对具有缺失值的归一化属性i，按照如下方式计算了两个案例X和Y之间的相似度：

$$\mathrm{sim}(x_i, y_i) = 0.5 \qquad \text{if } x_i \text{ or } y_i \text{ is unknown} \tag{14}$$

Surma and Vanhoof (1995) estimated the distance of missing values differently. For both unknown values, dist(x_i, y_i) = 0. If one attribute value is unknown and the other is known, then their distance is 1. Here attribute value is also normalized.

Summary

The similarity measures and related considerations discussed above are typical in the CBR literature. Some more complex measures are excluded because they are more or less context dependent. Another important way to improve measure performance is by weighting attributes according to their relative importance. Equation (8) describes one method to assign attribute weights. The method belongs to the IGNORANT category. Wettschereck and Aha (1995) compared five weighting methods. Overall, the basic principles for designing similarity measures are effectiveness and simplicity. The balance between them depends on the application.

OTHER SIMILARITY MEASURES

The notion of similarity plays a fundamental role in the knowledge and behavior theories. Besides the measures discussed in the previous section, the following general similarity measures can be potentially adopted for CBR applications.

Similarity Measure for Crisp Sets

Assume **X** and **Y** are crisp sets in n-dimensional space, $\mathbf{X} = (x_1, x_2, x_3, \ldots, x_n)$, $\mathbf{Y} = (y_1, y_2, y_3, \ldots, y_n)$, where x_i and y_i can only take 1 or 0. If x_i (or y_i) appears in **X** (or **Y**), it takes 1; otherwise, it takes 0. A list of typical similarity measures for crisp sets are as follows:

Inner product

$$|\mathbf{X} \cap \mathbf{Y}| \tag{15}$$

Dice coefficient

$$\frac{2|\mathbf{X} \cap \mathbf{Y}|}{|\mathbf{X}| + |\mathbf{Y}|} \tag{16}$$

Cosine coefficient

$$\frac{|\mathbf{X} \cap \mathbf{Y}|}{\sqrt{|\mathbf{X}|} \times \sqrt{|\mathbf{Y}|}} \tag{17}$$

Surma和Vanhoof（1995）对缺失值的距离进行了不同的估计。对于两个未知值，若某一属性值的取值为x_i，另一属性值的取值为y_i，且dist(x_i, y_i) = 0，则它们之间的距离为1。在此过程中，属性值同样经过了归一化处理。

Summary

上述讨论的相似性度量及相关考量在案例推理文献中较为常见。一些更为复杂的度量因在不同程度上依赖于具体情境而被排除在外。提升度量性能的另一重要途径是根据各属性的相对重要性为其赋予权重。式（8）描述了一种用于确定属性权重的方法，该方法属于“IGNORANT”类别。Wettschereck和Aha（1995）比较了五种权重分配方法。总体而言，设计相似性度量的基本原则是有效性和简洁性，二者之间的权衡则取决于具体应用。

其他相似性度量

相似性概念在知识与行为理论中起着 fundamental 的作用。除了前一节所讨论的度量方法外，以下几种通用的相似性度量也可能适用于案例推理（CBR）应用。

模糊集的相似性度量

设X和Y为n维空间中的清晰集，$X = (x_1, x_2, x_3, \cdots)$，其中各元素仅取0或1。若（或$x_i$），则$\mathbf{Y} = (y_1, y_2, y_3, \cdots)$，且当$x_i$出现在$\mathbf{X}$（或$\mathbf{Y}$）中时取1，否则取0。针对清晰集的一系列典型相似性度量如下：

内积

$$|\mathbf{X} \cap \mathbf{Y}| \tag{15}$$

骰子系数

$$\frac{2|\mathbf{X} \cap \mathbf{Y}|}{|\mathbf{X}| + |\mathbf{Y}|} \tag{16}$$

余弦系数

$$\frac{|\mathbf{X} \cap \mathbf{Y}|}{\sqrt{|\mathbf{X}|} \times \sqrt{|\mathbf{Y}|}} \tag{17}$$

Jaccard coefficient

$$\frac{|\mathbf{X} \cap \mathbf{Y}|}{|\mathbf{X}| + |\mathbf{Y}| - |\mathbf{X} \cap \mathbf{Y}|} \tag{18}$$

Overlap coefficient

$$\frac{|\mathbf{X} \cap \mathbf{Y}|}{\min(|\mathbf{X}|, |\mathbf{Y}|)} \tag{19}$$

where $|\mathbf{X}|$ is the number of elements in $\mathbf{X}$, $|\mathbf{Y}|$ is the number of elements in $\mathbf{Y}$, and $|\mathbf{X} \cap \mathbf{Y}|$ is the number of elements appearing jointly in $\mathbf{X}$ and $\mathbf{Y}$.

The inner product is, in principle, unbounded, while the other four coefficients are normalized. Other methods to measure the similarity between two crisp sets with numeric values are transformed measures such as

$$\mathrm{SIM}(\mathbf{X}, \mathbf{Y}) = \frac{2 \sum_i x_i y_i}{\sum x_i^2 + \sum y_i^2} \tag{20}$$

Similarity Measure for Fuzzy Sets

Since the first publication of fuzzy set theory by Zadeh (1965), great progress has been made in this area during the last three decades. One of the major issues is the similarity measure between two fuzzy sets (attribute values) (e.g., Zwick et al., 1987 ; Dubois et al., 1988 ; Miyamoto, 1990a, b ; Gu & Dubuisson, 1990 ; Pappis & Karacapilidis, 1993 ; Hyung et al., 1994 ; Chen et al., 1995 ; Wang et al., 1995 ; Liao & Zhang, 1996). The fuzzy similarity measures are usually based on the geometric model, the set-theoretic approach, and so on.

Measure Based on the Union and Intersection Operations

Assume $\mathbf{A}$ and $\mathbf{B}$ are fuzzy sets ; $\mathbf{a}$ and $\mathbf{b}$ are the vector representations of the fuzzy sets $\mathbf{A}$ and $\mathbf{B}$, respectively. The degree of similarity $M1_{A,B}$ is defined as follows:

$$M1_{A,B} = \frac{\sum_i (a_i \wedge b_i)}{\sum_i (a_i \vee b_i)} \tag{21}$$

where $\wedge$ ($\vee$) represents the min (max) operator, $\mathbf{a} = (a_1, a_2, \ldots, a_n)$, and $\mathbf{b} = (b_1, b_2, \ldots, b_n)$, $\mathbf{a} \in \mathbf{A}$, $\mathbf{b} \in \mathbf{B}$, n is the number of discrete values in the universe of discourse. $\mathbf{A}$ and $\mathbf{B}$ are considered to be approximately equal

雅卡尔系数

$$\frac{|\mathbf{X} \cap \mathbf{Y}|}{|\mathbf{X}| + |\mathbf{Y}| - |\mathbf{X} \cap \mathbf{Y}|} \tag{18}$$

重叠系数

$$\frac{|\mathbf{X} \cap \mathbf{Y}|}{\min(|\mathbf{X}|, |\mathbf{Y}|)} \tag{19}$$

其中，| X | 表示集合 X 中的元素个数，| Y | 表示集合 Y 中的元素个数，| X Ç Y | 表示同时属于 X 和 Y 的元素个数。

内积在原则上是无界的，而其余四个系数则经过归一化处理。用于度量两个具有数值属性的清晰集合之间相似性的其他方法包括诸如……之类的转换度量。

$$\mathrm{SIM}(\mathbf{X}, \mathbf{Y}) = \frac{2 \sum_i x_i y_i}{\sum x_i^2 + \sum y_i^2} \tag{20}$$

模糊集的相似度度量

自Zadeh于1965年首次提出模糊集理论以来，该领域在过去三十年中取得了长足进展。其中一个核心问题是如何度量两个模糊集（即属性值）之间的相似性（例如，Zwick等，1987；Dubois等，1988；Miyamoto，1990a、b；Gu与Dubuisson，1990；Pappis与Karacapilidis，1993；Hyung等，1994；Chen等，1995；Wang等，1995；Liao与Zhang，1996）。模糊相似性度量通常基于几何模型、集合论方法等。

Measure Based on the Union and Intersection Operations

设A和B为模糊集，a和b分别为模糊集A和B的向量表示。相似度M1(A,B)定义如下:

$$\mathrm{M1}_{\mathrm{A},\mathrm{B}} = \frac{\sum_i (a_i \wedge b_i)}{\sum_i (a_i \vee b_i)} \tag{21}$$

其中，∧ (∨) 表示最小（最大）运算符，……，且 a 5 （a_1, a_2, a），……，**a** ∈ **A**，**b** ∈ **B**，n 为论域 **b** 5 （b_1, b_2, b）中的离散值个数。**A** 和 **B** 被认为近似相等。

(denoted as $\mathbf{A} \sim \mathbf{B}$) if, for a given small nonnegative number ε, $M1_{A,B} \geq \varepsilon$. The number ε is said to be the proximity measure of **A** and **B**. This definition also applies to $M2_{A,B} \sim M6_{A,B}$ discussed below.

Measure Based on the Maximum Difference

The degree of similarity $M2_{A,B}$ between fuzzy sets **A** and **B** is defined as follows:

$$M2_{A,B} = 1 - \max_i(|a_i - b_i|) \tag{22}$$

Measure Based on the Difference and the Sum of Grades of Membership

The degree of similarity $M3_{A,B}$ between a pair of fuzzy sets **A** and **B** is calculated as

$$M3_{A,B} = 1 - \frac{\sum_i |a_i - b_i|}{\sum_i (a_i + b_i)} \tag{23}$$

or

$$M3_{A,B} = 1 - \frac{\sum_i |a_i \vee b_i - a_i \wedge b_i|}{\sum_i (a_i + b_i)} \tag{24}$$

Measure Based on the Geometric Distance Model

Consider that **a** and **b** are two points in n-dimensional space with components (a_i, b_i), $i = 1, 2, \ldots, n$, the degree of similarity $M4_{A,B}$ is defined based on the Minkowski r-metric ($r = 1$) as follows:

$$M4_{A,B} = 1 - \frac{\sum_i |a_i - b_i|}{n} \tag{25}$$

Measure Based on the Set-Theoretic Approach

Assume **A** and **B** are fuzzy sets in the universe of discourse **U**, the degree of similarity $M5_{A,B}$ between **A** and **B** is defined as follows:

$$M5_{A,B} = \sup_{x \in U} \mu_{A \cap B}(x) = \max_{x \in U} \min(\mu_A(x), \mu_B(x)) \tag{26}$$

where $\mu_A(x)$, $\mu_B(x)$, and $\mu_{A \cap B}(x)$ are the membership functions of **A**, **B** and the intersect of **A** and **B**, respectively. Note that when $\mathbf{A} = \mathbf{B}$, $M5_{A,B}$ does not always have value 1.

Measure Based on the Matching Function

In this method, the degree of similarity between two fuzzy sets **A** and **B** is defined as follows:

$$M6_{A,B} = \frac{\mathbf{a} \cdot \mathbf{b}}{\max(\mathbf{a} \cdot \mathbf{a}, \mathbf{b} \cdot \mathbf{b})} \tag{27}$$

(若对于给定的一个非负小数 ε，有 M1A ,B $ ε，则称其为 A ~B)。此时，称 ε 为 A 和 B 的接近度量。该定义同样适用于下文所讨论的情形。M2

A, B M6A,B

Measure Based on the Maximum Difference

模糊集A与B之间的相似度定义为M2A,B，其表达式如下：

$$M2_{A,B} = 1 - \max_i(|a_i - b_i|) \tag{22}$$

Measure Based on the Difference and the Sum of Grades of Membership

模糊集A与B之间相似度的度量M3(A, B)可按如下方式计算：

$$M3_{A,B} = 1 - \frac{\sum_i |a_i - b_i|}{\sum_i (a_i + b_i)} \tag{23}$$

或

$$M3_{A,B} = 1 - \frac{\sum_i |a_i \vee b_i - a_i \wedge b_i|}{\sum_i (a_i + b_i)} \tag{24}$$

Measure Based on the Geometric Distance Model

设 a 和 b 为 n 维空间中的两点，其各分量分别为 $i = 1, 2, \cdots, n$，则基于闵可夫斯基$_i r$-范数（当 $r = 1$ 时）的相似度定义为 (a, b)，记作 M4A,A, B：

$$M4_{A,B} = 1 - \frac{\sum_i |a_i - b_i|}{n} \tag{25}$$

Measure Based on the Set-Theoretic Approach

设A和B为论域U上的模糊集，A与B之间的相似度定义如下：M5

A, B

$$M5_{A,B} = \sup_{x \in U} \mu_{A \cap B}(x) = \max_{x \in U} \min(\mu_A(x), \mu_B(x)) \tag{26}$$

其中，和分别为A、B的隶属函数，而μA(x)、μB(x)以及μA ∩ B(x)则分别表示A与B的交集。需要注意的是，当A = B时，M5A,B并不总是取值为1。

Measure Based on the Matching Function

在该方法中，两个模糊集A与B之间的相似度定义如下：

$$M6_{A,B} = \frac{\mathbf{a} \cdot \mathbf{b}}{\max(\mathbf{a} \cdot \mathbf{a}, \mathbf{b} \cdot \mathbf{b})} \tag{27}$$

Other Similarity Measures

Wang et al. (1995) investigated another similarity measure inspired by the equivalence

$$\mathbf{A} = \mathbf{B} \Leftrightarrow \mathbf{A} \subseteq \mathbf{B} \cap \mathbf{B} \subseteq \mathbf{A}$$

This measure is defined as

$$E_{\wp}(\mathbf{A}, \mathbf{B}) = \min((\inf_{x \in U} \wp(\mathbf{A}(x), \mathbf{B}(x)), \inf_{x \in U} \wp(\mathbf{B}(x), \mathbf{A}(x))) \tag{28}$$

where $\wp$ is an implication operator, i.e., a $[0,1]^2 \rightarrow [0,1]$ mapping satisfying $\wp(0, 0) = \wp(0, 1) = \wp(1, 1) = 1$ and $\wp(1, 0) = 0$. The implication operator can be one of the following:

1. Gödel implication operator

$$\wp(x, y) = 1 \quad x \leq y$$
$$\wp(x, y) = y \quad \text{otherwise}$$

2. Goguen implication operator

$$\wp(x, y) = 1 \qquad x \leq y$$
$$\wp(x, y) = y/x \qquad \text{otherwise}$$

3. Luckasiewice implication operator

$$\wp(x, y) = \min(1, 1 - x + y)$$

Table 1 compares some similarity measures discussed above and summarizes their main properties. In the table, $\mathbf{A}^{\wedge}$ denotes the complement of $\mathbf{A}$,

Table 1 Main Properties of Fuzzy Similarity Measures

Properties	$M1_{A,B}$	$M2_{A,B}$	$M3_{A,B}$	$M4_{A,B}$	$M5_{A,B}$	$M6_{A,B}$
$M_{A,B} = M_{B,A}$	Y	Y	Y	Y	Y	Y
$\mathbf{A} = \mathbf{B} \Leftrightarrow M_{A,B} = 1$	Y	Y	Y	Y	N	Y
$\mathbf{A} \cap \mathbf{B} = 0 \Leftrightarrow M_{A,B} = 0$	Y	N	Y	N	Y	Y
$M_{A,A^{\wedge}} = 1 \Leftrightarrow \mathbf{A} = \mathbf{H}$	Y	Y	Y	Y	N	Y
$M_{A,A^{\wedge}} = 0 \Leftrightarrow \mathbf{A} = \mathrm{I}$ or $\mathbf{A} = 0$	Y	N	Y	Y	N	Y
$\mathbf{A} \sim \mathbf{B} \Rightarrow \mathbf{A} \cup \mathbf{C} \sim \mathbf{B} \cup \mathbf{C}$	N	Y	Y	N	N	N
$\mathbf{A} \sim \mathbf{B} \Rightarrow \mathbf{A} \cap \mathbf{C} \sim \mathbf{B} \cap \mathbf{C}$	N	Y	N	N	Y	Y
$\mathbf{A} \sim \mathbf{B} \Rightarrow \mathbf{A} \circ \mathbf{R} \sim \mathbf{B} \circ \mathbf{R}$	N	Y	N	N	Y	N
$\mathbf{R} \sim \mathbf{S} \Rightarrow \mathbf{A} \circ \mathbf{R} \sim \mathbf{A} \circ \mathbf{S}$	N	Y	N	N	N	N

Other Similarity Measures

Wang 等人（1995）研究了另一种受等价性启发的相似性度量。

$$\mathbf{A} = \mathbf{B} \Leftrightarrow \mathbf{A} \subseteq \mathbf{B} \cap \mathbf{B} \subseteq \mathbf{A}$$

该指标被定义为

$$E_{\wp}(\mathbf{A}, \mathbf{B}) = \min((\inf_{x \in U} \wp(\mathbf{A}(x), \mathbf{B}(x)), \inf_{x \in U} \wp(\mathbf{B}(x), \mathbf{A}(x))) \quad (28)$$

其中，$\wp$→为蕴涵算子，即一个从$[0,1]^2$到[0,1]的映射，满足：1) 1→1=1，且0→0=0。该蕴涵算子在(0,0)、(0,1)以及(1,1)处的取值可为以下形式之一：

1. 戈德尔蕴含算子

$$\wp(x, y) = 1 \quad x \le y$$
$$\wp(x, y) = y \quad \text{otherwise}$$

2. 戈根蕴涵算子

$$\wp(x, y) = 1 \quad x \le y$$
$$\wp(x, y) = y/x \quad \text{otherwise}$$

3. 卢卡谢维奇蕴含算子

$$\wp(x, y) = \min(1, 1 - x + y)$$

表1比较了上述讨论的若干相似性度量，并总结了它们的主要性质。在该表中，AÃ 表示 **A** 的补集。

表1 模糊相似度量的主要性质

Properties	$M1_{A,B}$	$M2_{A,B}$	$M3_{A,B}$	$M4_{A,B}$	$M5_{A,B}$	$M6_{A,B}$
$M_{A,B} = M_{B,A}$	Y	Y	Y	Y	Y	Y
$\mathbf{A} = \mathbf{B} \Leftrightarrow M_{A,B} = 1$	Y	Y	Y	Y	N	Y
$\mathbf{A} \cap \mathbf{B} = 0 \Leftrightarrow M_{A,B} = 0$	Y	N	Y	N	Y	Y
$M_{A,A^{\wedge}} = 1 \Leftrightarrow \mathbf{A} = \mathbf{H}$	Y	Y	Y	Y	N	Y
$M_{A,A^{\wedge}} = 0 \Leftrightarrow \mathbf{A} = \mathrm{I}$ or $\mathbf{A} = 0$	Y	N	Y	Y	N	Y
$\mathbf{A} \sim \mathbf{B} \Rightarrow \mathbf{A} \cup \mathbf{C} \sim \mathbf{B} \cup \mathbf{C}$	N	Y	Y	N	N	N
$\mathbf{A} \sim \mathbf{B} \Rightarrow \mathbf{A} \cap \mathbf{C} \sim \mathbf{B} \cap \mathbf{C}$	N	Y	N	N	Y	Y
$\mathbf{A} \sim \mathbf{B} \Rightarrow \mathbf{A}^{\circ}\mathbf{R} \sim \mathbf{B}^{\circ}\mathbf{R}$	N	Y	N	N	Y	N
$\mathbf{R} \sim \mathbf{S} \Rightarrow \mathbf{A}^{\circ}\mathbf{R} \sim \mathbf{A}^{\circ}\mathbf{S}$	N	Y	N	N	N	N

superscript zero denotes a max-min composition, **H** denotes 0.5 fuzzy set, **R** and **S** are any fuzzy sets, and $\cap$ and $\cup$ denote the intersection and union of two fuzzy sets, respectively.

HYBRID SIMILARITY MEASURE FOR FAILURE ANALYSIS

Zwick et al. (1987) and Liao and Zhang (1996) investigated the fuzzy similarity measures in various fields such as fuzzy expert system, fuzzy pattern recognition, and fuzzy query. Other authors (e.g., Willett, 1987 ; Salton, 1989 ; Miyamoto, 1990a, b ; Cain et al., 1991 ; Artymiuk et al., 1992 ; Gesu, 1994 ; Hyung et al., 1994 ; Kraslawski et al., 1995) discussed various applications of similarity measures. Because most of the traditional tools for formal modeling, reasoning, and computing are crisp, deterministic, and precise in nature, it is no surprise that the majority of proposed similarity measures in the CBR field are based on binary logic or crisp set, that is, the Yes-or-No type rather than the More-or-Less type. Kolodner (1993) and Watson and Marir (1994) stressed the importance of similarity function in CBR but did not discuss fuzzy similarity measure at all. After a careful survey, it was found that only Lee et al. (1995) used fuzzy set theory in CBR. However, linguistic attributes were not considered.

It has been recognized that fuzzy set theory can model reality more naturally and adequately. The fact that CBR systems used to handle human judgment, evaluation, reasoning, and decision cannot handle fuzzy concepts is a major constraint. To make the CBR system more powerful, the incorporation of fuzzy set theory is needed. In the following, the domain of failure analysis (Esaklul, 1990) is used to illustrate that applications with a mixture of crisp and fuzzy features are common in the real word. Subsequently, a hybrid similarity measure is proposed.

Characteristics of Failure Cases

Our failure cases were obtained from Scientific Testing Lab. Inc., a Louisiana company that has been conducting failure analysis business for more than 20 years. A CBR system is being developed to store failure analysis cases and to select most similar cases from the stored case base.

The data acquisition in failure analysis is an incremental process. During different stages of analysis, different actions and tests (e.g., chemical analyses, mechanical tests) are performed to identify the cause of a failure. Unknown or missing attribute values are filled in as the analysis progresses. Depending upon the failure, the total number of required tests vary. The CBR system might be run at different stages of the analysis.

上标零表示最大-最小复合运算，H 表示模糊集 0.5，R 和 S 为任意模糊集，Ç 和 È 分别表示两个模糊集的交集与并集。

用于故障分析的混合相似性度量

Zwick等人（1987年）以及Liao和Zhang（1996年）研究了模糊相似性度量在模糊专家系统、模糊模式识别和模糊查询等多个领域的应用。其他学者（如Willett，1987年；Salton，1989年；Miyamoto，1990a、1990b年；Cain等人，1991年；Artymiuk等人，1992年；Gesu，1994年；Hyung等人，1994年；Kraslawski等人，1995年）也探讨了相似性度量的多种应用。由于大多数用于形式建模、推理和计算的传统工具本质上是确定性的、精确的且非模糊的，因此在案例推理领域所提出的多数相似性度量均基于二值逻辑或经典集合论，即“是”或“否”类型的判断，而非“多一些”或“少一些”类型的判断，这并不令人意外。Kolodner（1993年）以及Watson和Marir（1994年）强调了相似性函数在案例推理中的重要性，但并未涉及模糊相似性度量。经仔细调研发现，仅有Lee等人（1995年）在案例推理中应用了模糊集理论，然而该研究并未考虑语言学属性。

人们已经认识到，模糊集理论能够更自然、更充分地刻画现实。然而，用于处理人类判断、评估、推理与决策的案例推理系统无法有效处理模糊概念，这一局限性构成了其主要制约因素。为提升案例推理系统的性能，有必要引入模糊集理论。下文以故障分析领域（Esaklul，1990）为例，说明在实际应用中同时包含精确与模糊特征的情形十分常见。随后，本文提出了一种混合型相似度度量方法。

故障案例的特征

我们的失效案例来源于Scienti®c Testing Lab. Inc.，这是一家位于路易斯安那州、从事失效分析业务超过20年的公司。目前正在开发一个基于案例的推理系统，用于存储失效分析案例，并从已存储的案例库中检索出最相似的案例。

失效分析中的数据采集是一个渐进式过程。在分析的不同阶段，会实施不同的操作与测试（如化学分析、力学试验），以确定失效原因。随着分析的推进，未知或缺失的属性值将逐步得到补充。根据具体的失效类型，所需测试的总数会有所差异。案例推理系统可能在分析的不同阶段被调用。

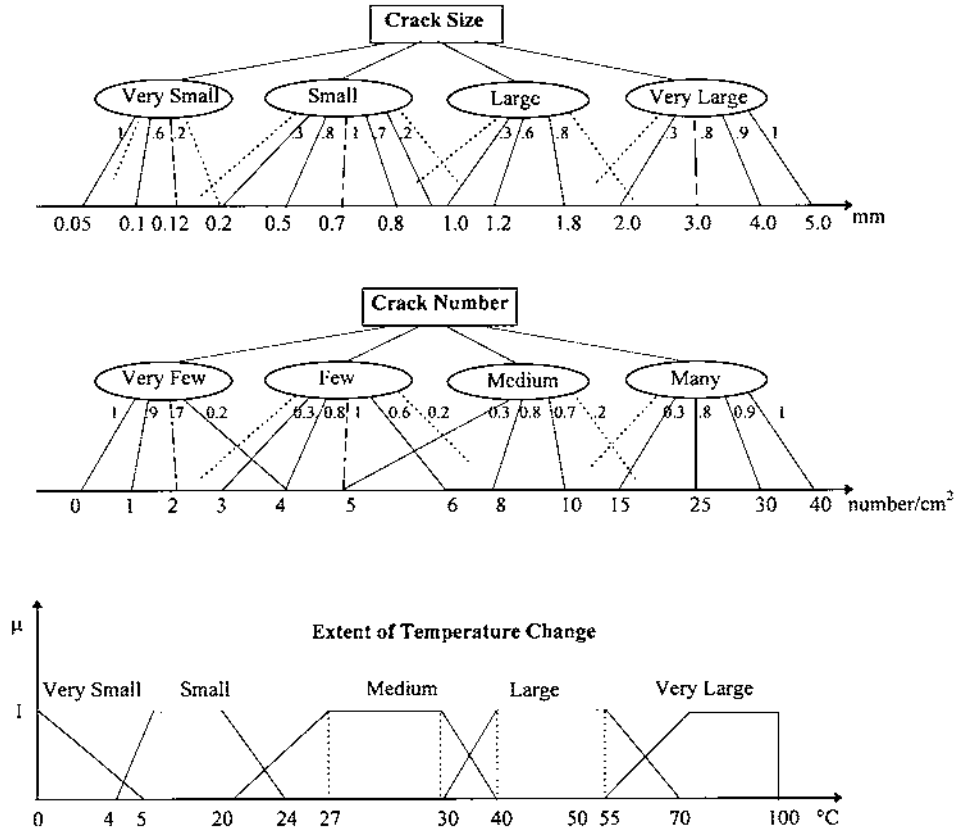


FIGURE 1. Examples of linguistic attributes.

The attributes used to represent failure cases could be associated with the following values:

- Crisp numeric values can be continuous or discrete. For example, the value of attribute "time in service" could be "5.5 years" or "4 years," depending upon the accuracy required.
- Crisp symbolic values are expressed as terms with definite meanings. For example, the value of the attribute "component form" is either "plate" or "pipe," etc. For improving user interactions and failure analysis process, a vocabulary of precise and consistent symbolic representation is needed.
- Fuzzy linguistic values can be considered as linguistic variables. The values of this kind of attribute form a term set. Each term in the term set corresponds to a fuzzy concept such as in Figure 1. Their definitions depend heavily on the application.

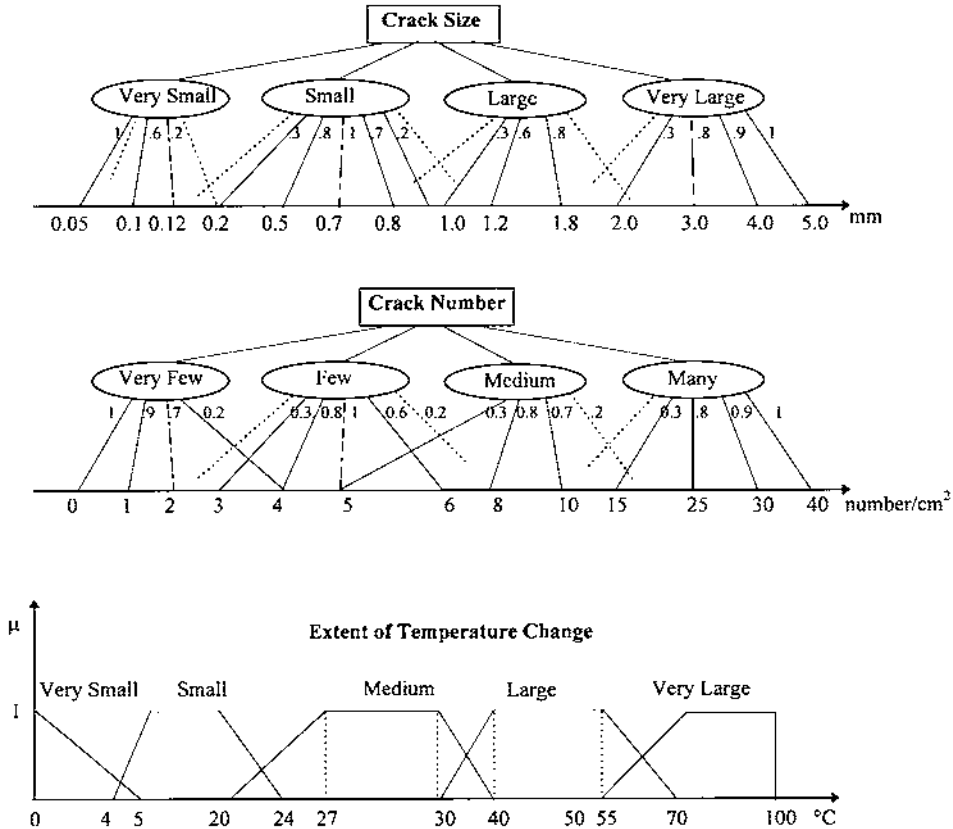


图1. 语言学属性示例。

用于表征故障案例的属性可与以下取值相关联:

数值可以是连续的，也可以是离散的。例如，属性“服务年限”的•清晰值可能是‘5.5年‘或4年‘，具体取决于所需的精度。符号值以具有明确含义的术语来表示。以•清晰值为例，属性“组件形式”的取值要么是‘板式‘，要么是‘管式‘，等等。为了改善用户交互并优化故障分析流程，需要建立一套精确且一致的符号化表示词汇体系。语言值可被视为语言变量；此类属性的•模糊值构成一个术语集，术语集中的每个术语对应于一个模糊概念，如图1所示。其定义在很大程度上依赖于具体应用。

- Fuzzy numbers or fuzzy intervals refer to an inexact number of interval with inexact bounds. For example, the pressure is about 220 kPa or the pressure ranges roughly between 100 and 150 kPa as shown in Figure 2.

Other characteristics involved in failure cases include the following:

- Attribute relevance: each case attribute contributes differently to the case matching process. The same attribute in different analysis stages might be of different relevance.
- Missing attributes: in the beginning of the analysis, the values of many attributes of the target case are unknown. Their values might not be known eventually if these attributes are considered not needed for the analysis. Missing attribute values are also generally present in the stored cases.
- Dominant attributes: some attributes are more important than others. This kind of attribute should be adequately weighted.
- Case stability: although each failure case might be unique, the causal relations for failures are relatively stable. In other words, the numbers of potential attributes contributing to each failure mode remain very much the same.

Hybrid Similarity Measure

In developing CBR systems for failure analysis or similar applications, how to measure fuzzy attributes is a major issue in designing a similarity measure. There are two issues involved: the similarity of two fuzzy concepts and the aggregation of all fuzzy attributes. The conventional similarity measures for CBR reviewed in the section above, Similarity Measures for CBR, do not provide a suitable solution for this problem.

Before applying fuzzy similarity measures, the membership functions of fuzzy concepts need to be defined. An LR representation of fuzzy sets (DuBois and Prader, 1980) is used in our study. Usually, fuzzy concepts or sets can be described by Gaussian membership functions, but the computation of Gaussian functions is complex. To make the computation easier,

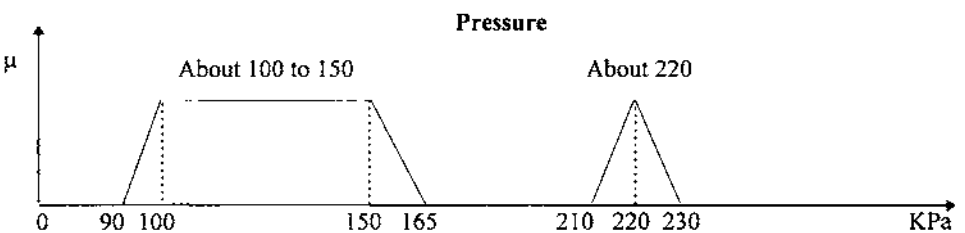


FIGURE 2. Examples of fuzzy number and interval attributes.

数值或模糊区间是指具有不精确边界的模糊区间●。例如，如图2所示，压力约为220 kPa，或大致介于100至150 kPa之间。

故障案例中涉及的其他特征包括以下几点：

相关性：每个案例属性对案例的贡献程度各不相同。同一属性在不同的分析阶段可能具有不同的相关性。属性：在分析初期，目标案例中许多缺失属性的取值尚不清楚。如果这些属性被认为对分析无必要，则其取值可能始终无法确定。存储的案例中通常也存在缺失的属性取值。属性：某些属性比其他属性更为重要。这类主导属性应给予适当的权重。稳定性：尽管每个故障案例可能具有独特性，但导致故障的因果关系相对稳定。换言之，对每种故障模式起作用的潜在属性数量基本保持不变。

混合相似度度量

在为故障分析或类似应用开发案例推理系统时，如何度量模糊属性是设计相似性度量的一个关键问题。该问题涉及两个方面：两个模糊概念之间的相似性以及所有模糊属性的聚合。上文“CBR中的相似性度量”一节所回顾的CBR传统相似性度量方法，并未为此问题提供合适的解决方案。

在应用模糊相似性度量之前，需要先确定模糊概念的隶属函数。本研究采用模糊集的LR表示法（DuBois和Prader，1980）。通常，模糊概念或模糊集可由高斯隶属函数来描述，但高斯函数的计算较为复杂。为简化计算，

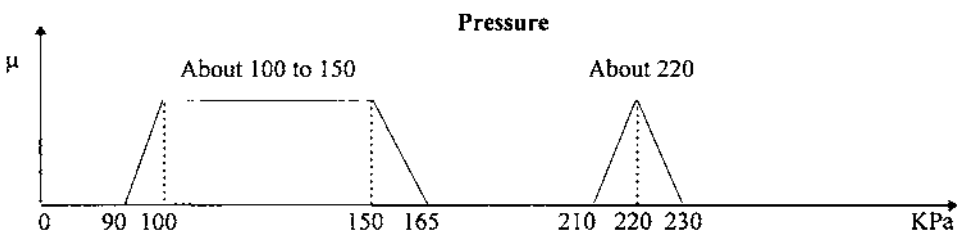


图2. 模糊数与区间属性的示例。

trapezoidal fuzzy sets are adopted to model the fuzzy attributes. The shape functions are as follows:

$$L(x) = R(x) = \max(0, 1 - x)$$

The membership function of the fuzzy set $\tilde{M}$ with parameter $\underline{m}$, $\bar{m}$, p, and q is

$$\mu_{\tilde{M}}(x) = \begin{cases} L\left(\dfrac{\underline{m} - x}{p}\right) & x \le \underline{m} \\ 1 & \underline{m} \le x \le \bar{m} \\ R\left(\dfrac{x - \bar{m}}{q}\right) & x \ge \bar{m} \end{cases}$$

For triangular fuzzy sets, $\underline{m} = \bar{m}$. The spreads p and q depend on the attributes. Taking the "extent of temperature change" attribute as an example of fuzzy linguistic attributes, the membership functions for "medium" and "large" are defined as follows:

$$\begin{aligned} \mu_{\tilde{M}}(x) &= L((27 - x)/7) && x \le 27 \\ \mu_{\tilde{M}}(x) &= 1 && 27 \le x \le 30 \\ \mu_{\tilde{M}}(x) &= R((x - 30)/10) && x \ge 30 \end{aligned} \tag{29}$$

$$\begin{aligned} \mu_{\tilde{L}}(x) &= L((40 - x)/10) && x \le 40 \\ \mu_{\tilde{L}}(x) &= 1 && 40 \le x \le 55 \\ \mu_{\tilde{L}}(x) &= R((x - 55)/15) && x \ge 55 \end{aligned} \tag{30}$$

For fuzzy linguistic attributes, the spreads p and q are assigned by domain experts. For fuzzy number and interval attributes, the values of p and q are considered to be $c\underline{m}$ and $c\bar{m}$, respectively. The default value of c is 0.1, but it can be changed by users. For example, for the attribute "pressure" with fuzzy interval value of "about 100–150," the default values of p and q are 10 and 15.

The section above, Other Similarity Measures, discussed several similarity functions and their main properties. It can be seen from Table 1 that $M5_{A,B}$ does not always hold the property $\mathbf{A} = \mathbf{B} \Leftrightarrow M_{A,B} = 1$. $M5_{A,B} = 1$ when $\mathbf{a} = (0.5, 1, 0.5)$ and $\mathbf{b} = (0.5, 1, 0.5)$. However, when $\mathbf{a} = (0.5, 0.5, 0.5)$ and $\mathbf{b} = (0.5, 0.5, 0.5)$, $M5_{A,B} = 0.5 \neq 1$. $M2_{A,B}$ and $M4_{A,B}$ do not always hold the property $\mathbf{A} \cap \mathbf{B} = 0 \Leftrightarrow M_{A,B} = 0$. For instance, if $\mathbf{A} = 1$ and $\mathbf{B} = 0$, then $\mathbf{A} \cap \mathbf{B} = 0$ and $M2_{A,B} = M4_{A,B} = 0$. But when $\mathbf{a} = (0.8, 0, 0)$ and $\mathbf{b} = (0, 0.9, 0)$, then $\mathbf{A} \cap \mathbf{B} = 0$, $M4_{A,B} = 1 - (0.8 + 0.9 + 0)/3 = 0.43 \neq 0$. $M2_{A,B} = 1 - \max(0.8, 0.9, 0) = 0.1 \neq 0$. So these similarity functions are not suitable for our use either. Accuracy and simplicity are the other major considerations. Commonly, discretizing the membership function yields lower

采用梯形模糊集来建模模糊属性。其隶属函数如下:

$$L(x) = R(x) = \max(0, 1 - x)$$

具有参数p、$\tilde{M}$ m、$\bar{m}$和q的模糊集的隶属函数为

$$\mu_{\tilde{M}}(x) = \begin{cases} L\left(\dfrac{\underline{m} - x}{p}\right) & x \le \underline{m} \\ 1 & \underline{m} \le x \le \bar{m} \\ R\left(\dfrac{x - \bar{m}}{q}\right) & x \ge \bar{m} \end{cases}$$

对于三角模糊集，其跨度p和q取决于属性m5$\bar{m}$。以“温度变化幅度”这一模糊语言属性为例，其中“中等”和“较大”的隶属函数分别定义如下:

$$\begin{aligned} \mu_{\tilde{M}}(x) &= L((27 - x)/7) && x \le 27 \\ \mu_{\tilde{M}}(x) &= 1 && 27 \le x \le 30 \\ \mu_{\tilde{M}}(x) &= R((x - 30)/10) && x \ge 30 \end{aligned} \tag{29}$$

$$\begin{aligned} \mu_{\tilde{L}}(x) &= L((40 - x)/10) && x \le 40 \\ \mu_{\tilde{L}}(x) &= 1 && 40 \le x \le 55 \\ \mu_{\tilde{L}}(x) &= R((x - 55)/15) && x \ge 55 \end{aligned} \tag{30}$$

对于模糊语言属性，区间宽度p和q由领域专家指定。对于模糊数及区间型属性，p和q的取值分别被视为 和 。c的默认值为cm $c\bar{m}$，即0.1，但用户可对其进行修改。例如，对于具有模糊区间值“约100～150”的属性“压力”，其p和q的默认值分别为10和15。

上文“其他相似性度量”部分讨论了若干相似性函数及其主要性质。由表1可知，性质M5（A, B）：A = B 当且仅当 MA,B = 1，并非总是成立。例如，当 a = (0.5, 1, 0.5)、b = (0.5, 1, 0.5) 时，MA,B = 1；然而，当 a = (0.5, 0.5, 0.5)、b = (0.5, 0.5, 0.5) 时，MA,B 并不总是等于0.5。此外，性质 M2(A, B) 和 M4(A, B) 也并非始终成立。例如，若 A = 1、B = 0，则 A ∩ B = 0 当且仅当 MA,B = 0。此时 A ∩ B = 0，但当 a = (0.8, 0, 0)、b = (0, 0.9, 0) 时，A ∩ B = 0，而 M4(A, B) = 1 - (0.8 × v26 + 0.9 × v27 + 0)/3 = 0.43 ≠ 0，M2(A, B) = 1 - max(0.8, 0) = 0.1 ≠ 0。因此，这些相似性函数也不适用于我们的应用场景。准确性和简洁性是另外两个重要的考量因素。通常，对隶属函数进行离散化处理会降低

accuracy. Our area ratio method has advantages in both accuracy and simplicity. The area ratio method calculates the similarity between two fuzzy values as the ratio of the overlapping area of two corresponding membership functions to the total area as follows:

$$\text{sim}(x_i, y_i) = A(x_i \cap y_i)/A(x_i \cup y_i) = A(x_i \cap y_i)/(A(x_i) + A(y_i) - A(x_i \cap y_i)) \tag{31}$$

where A represents the area of the corresponding membership function and $x_i \cap y_i$ is the intersection of two fuzzy sets. For fuzzy number and interval attributes, five types of intersection can be classified as depicted in Figure 3. For fuzzy linguistic attributes, only types (a), (b), and (c) apply. The following algorithm (FSM) is developed to calculate the fuzzy similarity $\text{sim}(x_i, y_i)$ for these types.

Algorithm FSM // Similarity measure for two fuzzy sets x_i and y_i.

(1) $c_{xi} \leftarrow (\underline{m}_{xi} + \bar{m}_{xi})/2$, $c_{yi} \leftarrow (\underline{m}_{yi} + \bar{m}_{yi})/2$ // calculate centers of two fuzzy sets.
(2) if $c_{xi} > c_{yi}$, then exchange x_i and y_i, endif
(3) $x_i^* \leftarrow (q_{xi}\underline{m}_{yi} + p_{yi}\bar{m}_{xi})/(p_{yi} + q_{xi})$, $y_i^* \leftarrow 1 - (x_i^* - \bar{m}_{xi})/q_{xi}$ // calculate intersection point (x_i^*, y_i^*).
(4) if $y_i^* \le 0$, then $A(x_i \cap y_i) \leftarrow 0$, $\text{sim}(x_i, y_i) \leftarrow 0$ // it belongs to type (a).
(5) else $A(x_i) \leftarrow (2\bar{m}_{xi} + q_{xi} - 2\underline{m}_{xi} + p_{xi})/2$, $A(y_i) \leftarrow (2\bar{m}_{yi} + q_{yi} - 2\underline{m}_{yi} + p_{yi})/2$ // calculate $A(x_i)$, $A(y_i)$.
(6) if $0 < y_i^* < 1$, then $A(x_i \cap y_i) \leftarrow ((\bar{m}_{xi} + q_{xi} - \underline{m}_{yi} + p_{yi})y_i^*)/2$ // it belongs to type (b).
(7) else // it belongs to type (c), (d), or (e).
(8) if $\bar{m}_{xi} + q_{xi} < \bar{m}_{yi} + q_{yi}$ and $\underline{m}_{xi} - p_{xi} < \underline{m}_{yi} - p_{yi}$, then // it belongs to type (c).
(9) $A(x_i \cap y_i) \leftarrow (2\bar{m}_{xi} + q_{xi} - 2\underline{m}_{yi} + p_{yi})/2$

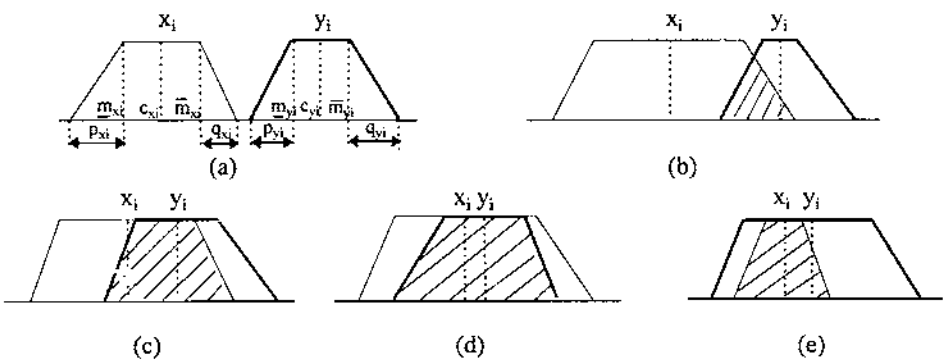


FIGURE 3. Different similarity types of two fuzzy sets x_i and y_i.

准确性。我们的面积比值法在准确性和简便性两方面均具有优势。该方法通过计算两个模糊值之间相似度的方式，即以两个相应隶属函数的重叠面积与总面积之比来表征，具体如下：

$$\mathrm{sim}(x_i, y_i) = A(x_i \cap y_i)/A(x_i \cup y_i) = A(x_i \cap y_i)/(A(x_i) + A(y_i) - A(x_i \cap y_i)) \tag{31}$$

其中，A 表示相应隶属函数的面积，即两个模糊集的交集。对于模糊数和区间属性 $x \cap y$，其交集可分为如图3所示的若干类型。对于模糊语言属性，仅适用类型(a)、(b)和(c)。为计算这些类型的模糊相似度 sim(x, y)，开发了如下算法（FSM）。

算法 FSM // 用于计算两个模糊集相似度的度量，记为 x y。

$x_i \leftarrow \underline{m}_{xi}$ x_i $y_i \leftarrow \underline{m}_{yi}$ y_i // 计算两个模糊集合的中心：(1) c $(m + \bar{m})/2$，c $(m + \bar{m})/2$。(2) 如果，则交换并结束；c，c，x_i y_i y。// 计算交点：(3) x^* $(q \quad m + p \quad \bar{m})/(p + q)$，$y^* 1 - (x^* - \bar{m})/q$，交点坐标为 (x^*, y^*)。(4) 如果，则属于类型(a)：$y^* \neq 0$，$A(x \cap y) = 0$，$\mathrm{sim}(x, y) = 0$。(5) 否则，$A(x)$ $(2 + \bar{m} + q^2 - 2m + p)/2$，$A(y)$ $(2 + \bar{m} \pm q^2 - 2m$，计算 $+ p)/2$ 和 $A(x)$，$A(y)$。(6) 如果，则属于类型(b)：$y^* = 0$，$A(x \cap y)$ $((\bar{m} + q = (m + p)y^* \times))/2$。(7) 否则，属于类型(c)、(d)或(e)。(8) 如果且满足条件，则属于类型(c)：$\bar{m} + q$，$\bar{m} + q$ $m - p$，$m - p$。(9) $A(x \cap y)$ $(2 + \bar{m} + q^2 - 2m + p)/2$。

x_i x_i y_i y_i $\underline{m}_{xi}$ x_i $\underline{m}_{yi}$ y_i

i $i \leftarrow$ x_i x_i $\underline{m}_{yi}$ y_i

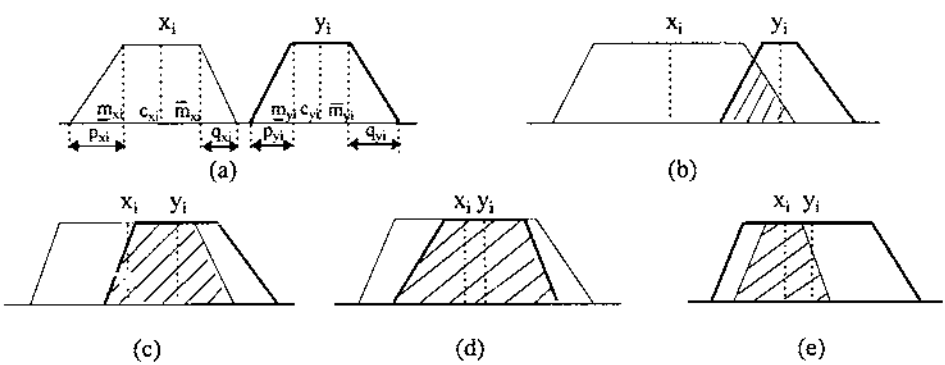


图3. 两个模糊集的多种相似性类型及其示例。 i i

(10) else $A(x_i \cap y_i) \leftarrow \min(A(x_i), A(y_i))$ // it belongs to type (d) or (e).
(11) endif
(12) endif
(13) $\text{sim}(x_i, y_i) \leftarrow A(x_i \cap y_i)/(A(x_i) + A(y_i) - A(x_i \cap y_i))$
(14) endif
(15) end_of_FSM.

In our study, the similarities of two fuzzy linguistic attributes for all possible combinations of fuzzy sets are calculated and stored in the system in advance. When matching cases, these values are retrieved from the database if needed to save computing time. This is sensible because all fuzzy sets are predefined for each fuzzy linguistic attribute. For the similarity between two fuzzy number or interval attributes, the distance between centers c_{xi} and c_{yi} is also considered, and the similarity is computed when two cases are matched. The weighted average operator is used to aggregate the similarities of all attributes.

In summary, a hybrid similarity measure proposed for matching two cases **X** and **Y** is

$$\text{SIM}(\mathbf{X}, \mathbf{Y}) = \sum_i w_i(\varepsilon_{i_1} \text{ sim1}_{\text{FNI}}(x_i, y_i) + \varepsilon_{i_2} \text{ sim2}_{\text{FNI}}(x_i, y_i)) + \sum_j w_j \text{ sim}_{\text{FL}}(x_j, y_j) + \sum_k w_k \text{ sim}_{\text{CS}}(x_k, y_k) + \sum_l w_l \text{ sim}_{\text{CN}}(x_l, y_l) \quad (32)$$

where Eq. (31) is used to calculate $\text{sim1}_{\text{FNI}}(x_i, y_i)$ and $\text{sim}_{\text{FL}}(x_j, y_j)$, and

$$\text{sim2}_{\text{FNI}}(x_i, y_i) = 1 - \text{dist}(c_{xi}, c_{yi}) = 1 - |c_{xi} - c_{yi}|/|\max_i - \min_i|$$
$$\text{sim}_{\text{CS}}(x_k, y_k) = 1 \qquad x_k = y_k$$
$$\text{sim}_{\text{CS}}(x_k, y_k) = 0 \qquad \text{otherwise}$$
$$\text{sim}_{\text{CN}}(x_l, y_l) = 1 - \text{dist}(x_l, y_l) = 1 - |x_l - y_l|/|\max_l - \min_l|$$

where w_i, w_j, w_k, and w_l are the weights for each attribute and $\Sigma w_i + \Sigma w_j + \Sigma w_k + \Sigma w_l = 1$; $i = 1, \ldots, n1$; $j = n1 + 1, \ldots, n1 + n2$; $k = n1 + n2 + 1, \ldots, n1 + n2 + n3$; $l = n1 + n2 + n3 + 1, \ldots n$; $n1$, $n2$, $n3$, and n are the number of fuzzy number and interval, fuzzy linguistic, crisp symbolic, and total attributes, respectively. The default values of ε_{i_1} and ε_{i_2} are taken as follows:

$$\varepsilon_{i_1} = \min\left[\max_{v_{ij}, j=1,\ldots,t}(\max(v_{ij}) - \min(v_{ij}))\Big/\left(\max_{v_{ij}, j=1,\ldots,t} - \min_{v_{ij}, j=1,\ldots,t}(v_{ij})\right), 0.9\right]$$

$$\varepsilon_{i_2} = 1 - \varepsilon_{i_1}$$

(10) 否则 // 其属于类型 (d) 或 $A(x \cap y) \subseteq \min(A(x), A(y))$ (e)
。
(11) 否则
(12) 如果（13）sim(*x*, *y*) 成立，则 $A(x \cap y) / (A(x) + A(y) - A(x \cap y))$ (14) 。否则，结束状态机的执行。

在本研究中，针对所有可能的模糊集组合，预先计算并存储两种模糊语言属性之间的相似度。在案例匹配时，若有必要，可直接从数据库中调取这些值以节省计算时间。这一做法具有合理性，因为每种模糊语言属性的所有模糊集均已预先定义。对于两个模糊数或区间属性之间的相似度，还考虑了中心点之间的距离，并在两个案例进行匹配时计算该相似度。最后，采用加权平均算子对所有属性的相似度进行综合聚合。

综上所述，为匹配两个案例X和Y而提出的混合相似性度量是

$$\mathrm{SIM}(X, Y) = \sum_i w_i \left(\varepsilon_{i1} \mathrm{sim}_{\mathrm{FNI}}(x_i, y_i) + \varepsilon_{i2} \sum_{\mathrm{FNI}} (x_i, y_i)\right) + \sum_j w_j \mathrm{sim}_{\mathrm{FL}}(x_j, y_j) + \sum_k w_k \mathrm{sim}_{\mathrm{CS}}(x_k, y_k) + \sum_1 w_1 \mathrm{sim}_{\mathrm{CN}}(x_1, y_1) \quad (32)$$

其中，式（31）用于计算 和 以及 $\mathrm{sim1}_{\mathrm{FNI}}(x_i, y_i)$、$\mathrm{sim}_{\mathrm{FL}}(x_j, y_j)$。

$$\mathrm{sim2}_{\mathrm{FNI}}(x_i, y_i) = 1 - \mathrm{dist}(c_{xi}, c_{yi}) = 1 - |c_{xi} - c_{yi}| / |\max_i - \min_i|$$

$$\mathrm{sim}_{\mathrm{CS}}(x_k, y_k) = 1 \quad x_k = y_k$$

$$\mathrm{sim}_{\mathrm{CS}}(x_k, y_k) = 0 \quad \text{otherwise}$$

$$\mathrm{sim}_{\mathrm{CN}}(x_l, y_l) = 1 - \mathrm{dist}(x_l, y_l) = 1 - |x_l - y_l| / |\max_l - \min_l|$$

其中，和分别为各属性的权重；w_i、w_j、w_k、w_l、w_1 +、i分别取值为1, j…，k1；j等于n1加上+1，…，再加上n1加上+ n2；k等于n1加上+ n2，再与w_1 +、w_1 +相加，结果为1；+取值为1，…，n1加上+ n2再加上+ n3；l等于n1、+ n2、+ n3之和，且取值范围为1，…，n；n1、n2、n3以及n分别表示模糊数与区间型属性、模糊语言型属性、清晰符号型属性以及总属性的数量。和的默认值分别取为ε_i1和ε_i2，具体如下：

$$\varepsilon_{i1} = \min\left[\max_{v_{ij}, j=1,\ldots,t} (\max(v_{ij}) - \min(v_{ij})) \Big/ (\max_{v_{ij}, j=1,\ldots,t} - \min_{v_{ij}, j=1,\ldots,t} (v_{ij})), 0.9\right]$$

$$\varepsilon_{i2} = 1 - \varepsilon_{i1}$$

where v_{ij} denotes the value of the ith attribute, and t is the total number of values having occurred in the ith attribute, which equals to the number of cases in the case base plus 1. The related parameters such as ε_{i_1}, $\max(v_{ij})$, and $\min(v_{ij})$ are also saved in the case base as facets of the ith attribute and can be updated dynamically as new cases are added. Note that the minimum value of ε_{i_2} is 0.1 to assure that the similarity between two values without intersection can be measured. In Eq. (32), the higher the weight, the more important the attribute is. In our study the weights are assigned by experts using Saaty's eigenvector method.

The matching score for the attributes with missing values is estimated as follows. In the initial analysis stages, the attributes with missing values are ignored so that more similar cases can be retrieved. They are then used to identify the most important missing data (attribute). The most important missing data could be found on the basis of an information theoretical criterion or simply the most common one shared by the retrieved cases. The identification of this attribute provides guidance for the incremental failure analysis process. In the last analysis stage, for symbolic and linguistic attributes, the matching score for missing attributes is estimated to be $1/N_i$ regardless if both or one value is unknown (N_i is the number of possible values of the ith attributes). For crisp numeric, fuzzy number, and interval attributes, the matching score is estimated to be 0.5 regardless if both or one value is unknown.

Dominant attributes can be considered as hard constraints to filter out unsuitable cases or placed at higher level nodes in the hierarchical indexing tree.

Test Example

The following example shows how the proposed hybrid measure works. Two cases **X**(case1) and **Y**(case2) of failure analysis are used; $n1$, $n2$, $n3$, and n are 2, 3, 7, and 16, respectively. The type and weight for each attribute are listed in Table 2, where CS, CN, FL, FN, and FI denote crisp symbolic, crisp numeric, fuzzy linguistic, fuzzy number, and fuzzy interval types, respectively. For the CS attributes, taking "form" as an example, the weighted similarity for this attribute is $0.1 \times 1 = 0.1$ because the values of **X** and **Y** both are "pipe" and the weight is 0.1. For the attribute "flow rate," its type is FN, x_i is about 115, and y_i is about 80. Applying the FSM algorithm, $c_{xi} = 115$ and $c_{yi} = 80$. Because of $c_{xi} > c_{yi}$, exchange x_i and y_i. Thus $c_{xi} = 80$ and $c_{yi} = 115$. Also,

$$\underline{m}_{xi} = \bar{m}_{xi} = 80 \qquad \underline{m}_{yi} = \bar{m}_{yi} = 115 \qquad p_{xi} = q_{xi} = 8 \qquad p_{yi} = q_{yi} = 11.5$$

$$x_i^* = (8 \times 115 \times 11.5 \times 80)/11.5 + 8) = 94.3590$$

$$y_i^* = 1 - (94.3590 - 80)/8 = -0.7949$$

其中，i_i表示第i个属性的取值，而t为在第i个属性中出现的v类取值的总数，该值等于案例库中的案例数加1。相关的参数，如ε、max(v)以及，也被作为第i个属性的特征保存在案例库中，并且随着新案例的加入，min(v)可动态更新。需要注意的是，的最小值设为0.1，以确保能够对两个无交集的取值之间的相似度进行度量。在式（32）中，权重越高，该属性的重要性越大。在本研究中，权重由专家采用Saaty的特征向量法确定。

对于具有缺失值的属性，其匹配得分按如下方式估计。在初始分析阶段，为便于检索出更多相似案例，暂不考虑这些具有缺失值的属性；随后，利用这些属性确定最重要的缺失数据（即缺失的属性）。最重要的缺失数据可根据信息论准则加以识别，或简单地选取被检索到的案例中出现频率最高的取值。该属性的识别为增量式故障分析过程提供了指导。在最后的分析阶段，对于符号型和语言型属性，无论两个值均未知还是仅一个值未知，其缺失属性的匹配得分均估计为1/N，其中N表示第i个属性的可能取值个数N。对于精确数值型、模糊数型及区间型属性，无论两个值均未知还是仅一个值未知，其匹配得分均估计为0.5。

主导属性可被视为硬约束，用于过滤掉不合适的案例，或置于层次化索引树的较高层级节点。

测试示例

以下示例展示了所提出的混合度量方法的运作机制。以两个故障分析案例X（case1）和Y（case2）为例，其中n1、n2、n3和n的取值分别为2、3、7和16。各属性的类型及其权重见表2，其中CS、CN、FL、FN和FI分别表示确定性符号型、确定性数值型、模糊语言型、模糊数值型和模糊区间型。对于CS类型的属性，以“形式”为例，由于X和Y的取值均为“管道”，且该属性的权重为0.1，因此其加权相似度为0.1×1=0.1。对于属性“流量”，其类型为FN，X的取值约为115，Y的取值约为80。应用模糊相似度度量算法，可得x i y，并且由于交换性，有c=115，c=80。c。c，x y。c=80，同时c=115。

$$\underline{m}_{xi} = \bar{m}_{xi} = 80 \qquad \underline{m}_{yi} = \bar{m}_{yi} = 115 \qquad p_{xi} = q_{xi} = 8 \qquad p_{yi} = q_{yi} = 11.5$$

$$x_i^* = (8 \times 115 \times 11.5 \times 80)/11.5 + 8) = 94.3590$$

$$y_i^* = 1 - (94.3590 - 80)/8 = -0.7949$$

Table 2 Example with Hybrid Similarity Measure

Attribute	Type	W_i	X	Y	Similarity
ID	CS	0	00V1143	00V1185	0
Form	CS	0.1	pipe	pipe	0.1
Crack size	FL	0.08	small	large	0.0026
Crack number	FL	0.07	many	medium	0.0052
Temperature change	FL	0.06	large	medium	0.0041
Pressure	FI	0.06	100–150 kPa	130–200 kPa	0.0413
Material	CS	0.1	carbon steel	carbon steel	0.1
Service time	CN	0.01	5.5Y	4Y	0.0097
Chemical condition	CS	0.1	acid	acid	0.1
Surface debris	CS	0.09	rust	rust	0.09
Symptom	CS	0.1	leaking	leaking	0.1
Flow rate	FN	0.02	115 mm/s	80 mm/s	0.0112
Hardness	CN	0.05	159 HK	197 HK	0.0481
Tensile strength	CN	0.05	519 MPa	508 MPa	0.0495
Fracture toughness K_{IC}	CN	0.04	60 MPa m	66 Mpa m	0.0395
Service type	CS	0.07	continuous	continuous	0.07
Total		1			0.7712

So it belongs to type (a) in Figure 3 and $\text{sim1}_{\text{FNI}}(x_i, y_i) = 0$, $\text{sim2}_{\text{FNI}}(x_i, y_i) = 1 - |80 - 115|/(500 - 0) = 1 - 0.07 = 0.93$. Say retrieved ε_{i_1} is 0.4, which is not modified by the new case, so $\varepsilon_{i_2} = 1 - 0.4 = 0.6$. Therefore the weighted similarity is $0.02 \times (0.4 \times 0 + 0.6 \times 0.93) = 0.0112$.

Taking "pressure" as an example of the FI type attribute, x_i is "about 100–150" and y_i is "about 130–200." According to the algorithm, c_{xi} and c_{yi} are 125 and 165, respectively, and

$$\underline{m}_{xi} = 100 \quad \bar{m}_{xi} = 150 \quad \underline{m}_{yi} = 130 \quad \bar{m}_{yi} = 200$$

$$p_{xi} = 10 \quad q_{xi} = 15 \quad p_{yi} = 13 \quad q_{yi} = 20$$

$$x_i = (15 \times 130 + 13 \times 150)/(13 + 15) = 139.2857$$

$$y_i^* = 1 - (139.2857 - 150)/15 = 1.7143$$

$$A(x_i) = (2\bar{m}_{xi} + q_{xi} - 2\underline{m}_{xi} + p_{xi})/2 = (2 \times 150 + 15 - 2 \times 100 + 10)/2 = 62.5$$

$$A(y_i) = (2\bar{m}_{yi} + q_{yi} - 2\underline{m}_{yi} + p_{yi})/2 = (2 \times 200 + 20 - 2 \times 130 + 13)/2 = 86.5$$

Because

$$\bar{m}_{xi} + q_{xi} = 150 + 15 = 165 < \bar{m}_{yi} + q_{yi} = 200 + 20 = 220$$

$$\underline{m}_{xi} - p_{xi} = 100 - 10 = 90 < \underline{m}_{yi} - p_{yi} = 130 - 13 = 117$$

表2 基于混合相似性度量的示例

Attribute	Type	W_i	X	Y	Similarity
ID	CS	0	00V1143	00V1185	0
Form	CS	0.1	pipe	pipe	0.1
Crack size	FL	0.08	small	large	0.0026
Crack number	FL	0.07	many	medium	0.0052
Temperature change	FL	0.06	large	medium	0.0041
Pressure	FI	0.06	100–150 kPa	130–200 kPa	0.0413
Material	CS	0.1	carbon steel	carbon steel	0.1
Service time	CN	0.01	5.5Y	4Y	0.0097
Chemical condition	CS	0.1	acid	acid	0.1
Surface debris	CS	0.09	rust	rust	0.09
Symptom	CS	0.1	leaking	leaking	0.1
Flow rate	FN	0.02	115 mm/s	80 mm/s	0.0112
Hardness	CN	0.05	159 HK	197 HK	0.0481
Tensile strength	CN	0.05	519 MPa	508 MPa	0.0495
Fracture toughness K_{IC}	CN	0.04	60 MPa m	66 Mpa m	0.0395
Service type	CS	0.07	continuous	continuous	0.07
Total		1			0.7712

因此，它属于图3中的类型(a)，且$sim1_{FNI}(x, y) = 0$，$sim2_{FNI}(x_i, y_i) = 1 - |80 - 115|/(500 - 0) = 1 - 0.07 = 0.93$。假设检索结果为0.4，该值未受新案例的影响，因此加权后的$\varepsilon_i = 1 - 0.4 = 0.6$。相似度为0.02（0.4 0 + 0.6 0.93）× 0.0112。

以“压力”这一FI类属性为例，其取值范围为“约100–150”，且“约为130–200”。根据该算法，y_i和c_i的值分别为x_{mi}125和y_{mi}65。

$$\underline{m}_{xi} = 100 \quad \bar{m}_{xi} = 150 \quad \underline{m}_{yi} = 130 \quad \bar{m}_{yi} = 200$$

$$p_{xi} = 10 \quad q_{xi} = 15 \quad p_{yi} = 13 \quad q_{yi} = 20$$

$$x_i = (15 \times 130 + 13 \times 150)/(13 + 15) = 139.2857$$

$$y_i^* = 1 - (139.2857 - 150)/15 = 1.7143$$

$$A(x_i) = (2\bar{m}_{xi} + q_{xi} - 2\underline{m}_{xi} + p_{xi})/2 = (2 \times 150 + 15 - 2 \times 100 + 10)/2 = 62.5$$

$$A(y_i) = (2\bar{m}_{yi} + q_{yi} - 2\underline{m}_{yi} + p_{yi})/2 = (2 \times 200 + 20 - 2 \times 130 + 13)/2 = 86.5$$

因为

$$\bar{m}_{xi} + q_{xi} = 150 + 15 = 165 < \bar{m}_{yi} + q_{yi} = 200 + 20 = 220$$

$$\underline{m}_{xi} - p_{xi} = 100 - 10 = 90 < \underline{m}_{yi} - p_{yi} = 130 - 13 = 117$$

it belongs to type (c) and

$$A(x_i \cap y_i) = (2\bar{m}_{xi} + q_{xi} - 2\underline{m}_{yi} + p_{yi})/2$$
$$= (2 \times 150 + 15 - 2 \times 130 + 13)/2 = 34$$
$$\text{sim1}_{\text{FNI}}(x_i, y_i) = A(x_i \cap y_i)/(A(x_i) + A(y_i) - A(x_i \cap y_i))$$
$$= 34/(62.5 + 86.5 - 34) = 0.2957$$
$$\text{sim2}_{\text{FNI}}(x_i, y_i) = 1 - |125 - 165|/(320 - 0) = 1 - 0.125 = 0.875$$

Say retrieved ε_{i_1} is 0.25. Because of $(200 + 20) - (130 - 13) = 103$ and $103/320 = 0.3219 > 0.25$, ε_{i_1} is replaced with 0.3219 and $\varepsilon_{i_2} = 1 - 0.3219 = 0.6781$. So the weighted similarity for this attribute is $0.06 \times (0.3219 \times 0.2957 + 0.6781 \times 0.875) = 0.0413$.

As an example of the FL attributes, the values of "extent of temperature change" in **X** and **Y** are "large" and "medium." The similarity value of "large" and "medium" is 0.0685, which is retrieved from the database. The 0.0685 value was precalculated according to the FSM algorithm based on the membership functions given in Eqs. (29) and (30). Therefore the weighted similarity for this attribute is $0.06 \times 0.0685 = 0.0041$. An example of CN attributes is "service time," the values of the attribute in **X** and **Y** are "5.5" and "4," the corresponding weighted similarity is $0.01 \times (1 - |5.5 - 4|/(50 - 0)) = 0.01 \times 0.97 = 0.0097$. Thus the total similarity of **X** and **Y**, according to Eq. (32), is 0.7712.

Table 2 demonstrates that, without the fuzzy measure, the similarity for three FL types attributes cannot be measured and the related similarities $0.0026 + 0.0052 + 0.0041 = 0.0119$ will be lost. For the FI and FN type attributes, only the distances between corresponding centers can be used to calculate the similarities. It is clear that the total similarity with a pure crisp similarity measure for the same cases, **X** and **Y**, cannot reflect the real similarity correctly.

CONCLUSION

We have reviewed various similarity measures in the literature and proposed a hybrid similarity measure for handling applications with a mixture of crisp and fuzzy features. Its application to failure analysis is illustrated. Results show that our hybrid measure can reflect the domain nature effectively. The more the fuzziness in the case attributes, the more the power of

它属于类型(c)，并且

$$A(x_i \cap y_i) = (2\bar{m}_{xi} + q_{xi} - 2\underline{m}_{yi} + p_{yi})/2$$

$$= (2 \times 150 + 15 - 2 \times 130 + 13)/2 = 34$$

$$\text{sim1}_{\text{FNI}}(x_i, y_i) = A(x_i \cap y_i)/(A(x_i) + A(y_i) - A(x_i \cap y_i))$$

$$= 34/(62.5 + 86.5 - 34) = 0.2957$$

$$\text{sim2}_{\text{FNI}}(x_i, y_i) = 1 - |125 - 165|/(320 - 0) = 1 - 0.125 = 0.875$$

设检索得到的v_{10}值为0.25。由于(200 × v10 + 20) ÷ (130 ÷ 13) = 5 × 10^3，且10^3 / v11 = 1320，因此v11的值被替换为0.3219，而v12 + v13 = 1 - 0.3219 = 0.6781。于是，该属性的加权相似度为0.06 × (0.3219 × 0.2957 + 0.6781 × 0.875) = 0.0413。

作为模糊逻辑属性的一个示例，X和Y中“温度变化幅度”的取值分别为“大”和“中”，而“大”与“中”的相似度为0.0685，该值系从数据库中检索所得。此0.0685的数值是依据式（29）和式（30）所给的隶属函数，按照模糊相似性度量（FSM）算法预先计算得到的。因此，该属性的加权相似度为0.06 × 0.0685 = 0.0041。作为冲突网络属性的一个示例，“服务时间”的取值在X和Y中分别为“5.5”和“4”，其对应的加权相似度为0.01 × [1 - |5.5 - 4| / (50 - 0)] = 0.01 × 0.97 = 0.0097。由此，根据式（32），X与Y的总相似度为0.7712。

表2表明，若不采用模糊测度，则无法对三种FL型属性的相似性进行度量，相关相似度0.0026 +、0.0052 +、0.0041 5、0.0119将被丢失。对于FI型和FN型属性，仅能利用对应中心之间的距离来计算其相似性。显然，对于同一案例X和Y，仅采用纯清晰相似性度量所得到的总相似度无法准确反映其真实相似程度。

结论

我们回顾了文献中提出的多种相似性度量方法，并提出了一种混合相似性度量，用于处理同时包含精确特征和模糊特征的应用场景。本文还对其在故障分析中的应用进行了说明。结果表明，该混合度量能够有效反映领域特性；案例属性中的模糊性程度越高，其表现能力越强。

the proposed measure. It is thus shown that combining fuzzy set theory with the conventional CBR system greatly enhances its capability.

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所提出的度量方法。由此表明，将模糊集理论与传统的案例推理系统相结合，可显著提升其性能。

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